

Homework 1

Due Friday Feb 27, 2026, 11 AM

General instructions:

- ☐ Submit all code, output and discussion as a Jupyter notebook.
 - ☐ Briefly describe how and why you collaborated and/or used generative AI for this work.
 - ☐ If you come across any good references, please include them in your document.
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1. Complete the PCA analysis and filtering of the Dow training data from Week 1 notebook. After filtering for linearized impurity data, briefly discuss (>150 words) how your dataset compares to Qin et al.
 2. Apply K-means and/or hierarchical clustering to the *unfiltered* Dow training data. Can we identify similar issues as revealed by PCA?
 3. Using the TEP process data (Refs. 1-3), apply unsupervised techniques to **one** of the fault data files (`d01.dat` -- `d21.dat`). Using the chosen techniques, can we identify the fault?

References

1. "The Tennessee Eastman Process: an Open Source Benchmark" <https://keepfloyding.github.io/categories/tep/>, Accessed Feb 20, 2026.
2. Rieth, Cory A.; Amsel, Ben D.; Tran, Randy; Cook, Maia B., 2017, "Additional Tennessee Eastman Process Simulation Data for Anomaly Detection Evaluation", <https://doi.org/10.7910/DVN/6C3JR1>, Harvard Dataverse, V1
3. "Tennessee Eastman Process CSV" <https://www.kaggle.com/datasets/afrniomelo/tep-csv>, Accessed Feb 20, 2026.
4. C. A. Rieth, B. D. Amsel, R. Tran and M. B. Cook, "Issues and advances in anomaly detection evaluation for joint human-automated systems", in *Advances in Human Factors in Robots and Unmanned Systems*, ed. J. Chen (Springer International Publishing, Cham, 2018), pp. 52-63. ([link](#))